Wireshark

Wireshark, a powerful network analysis tool, was formerly known as Ethereal. Its primary function is to capture network packets in real-time, presenting them in a human-readable format. Beyond merely displaying packet data, Wireshark offers an extensive array of features including filters, color-coding, and in-depth packet inspection capabilities. This tutorial is designed to provide a foundational understanding of how to capture packets, apply filters, and inspect network traffic using Wireshark. The tool is versatile and can be utilized for various purposes, such as investigating a program's network activity, analysing network traffic patterns, or troubleshooting network issues.

Obtaining Wireshark, You can obtain Wireshark for your Windows or Mac OS X system by downloading it from the official website. If you are working with a Linux or another UNIX-like system, you'll likely find Wireshark conveniently available in your system's package repositories. https://www.wireshark.org/download.html

It's important to note that many organizations have policies in place that prohibit the use of Wireshark and similar tools on their networks. Before using Wireshark in a professional environment, it is crucial to obtain the necessary permissions and approvals.

Capturing Packets (PLEASE CONDUCT THIS LAB AT HOME ON YOUR OWN NETWORK)

Once Wireshark has been downloaded and installed, launching the application allows you to initiate the packet capture process. Here's how:

1. Launch Wireshark on your computer.
2. Double-click the name of a network interface that you want to capture packets from. These network interfaces correspond to your computer's

Michael Hegarty

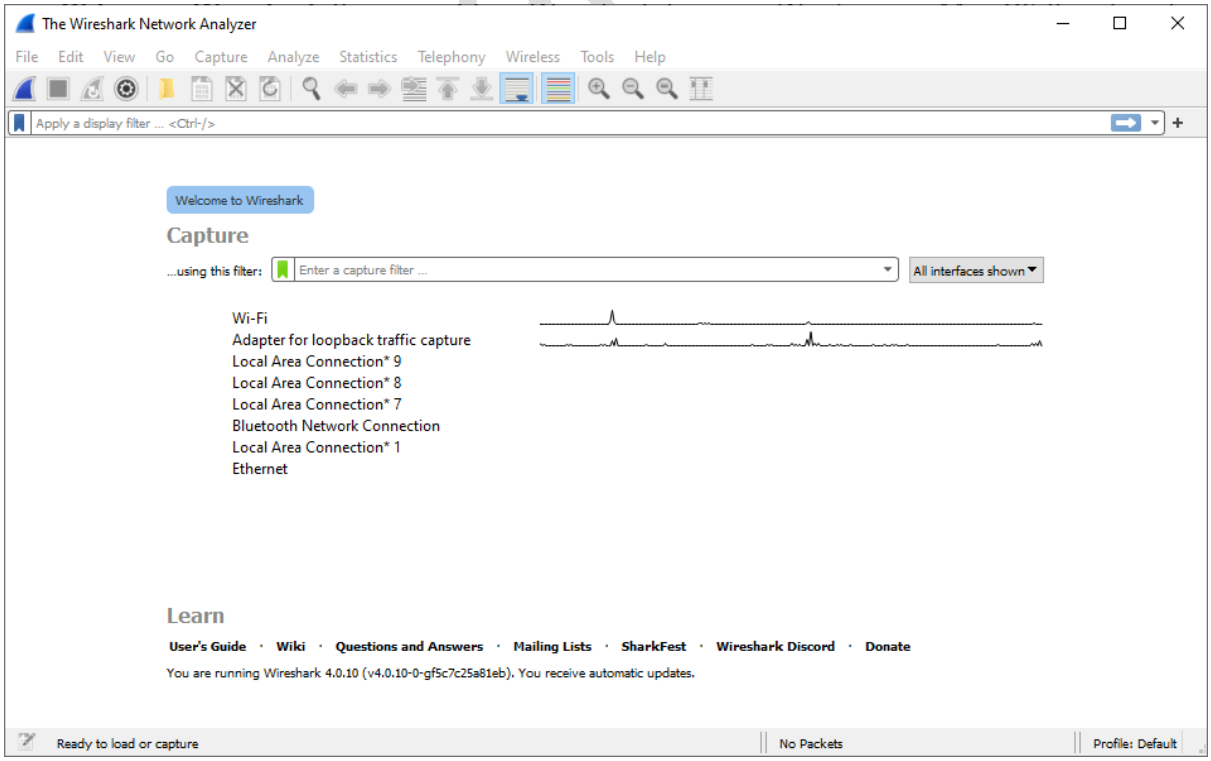
physical or virtual network connections. For instance, if you wish to monitor the traffic on your wireless network, select your wireless interface.

Note: The list of network interfaces can be found under the "Capture" menu.

3. At this point, Wireshark will start capturing packets on the chosen network interface in real-time.

While Wireshark offers a range of advanced features that can be configured for more specific capture needs, these advanced settings are not necessary for basic packet capture. For more complex tasks, you can explore these advanced features by navigating to "Capture" and selecting "Options." However, for beginners, capturing packets by double-clicking a network interface is sufficient to get started with the packet analysis process.

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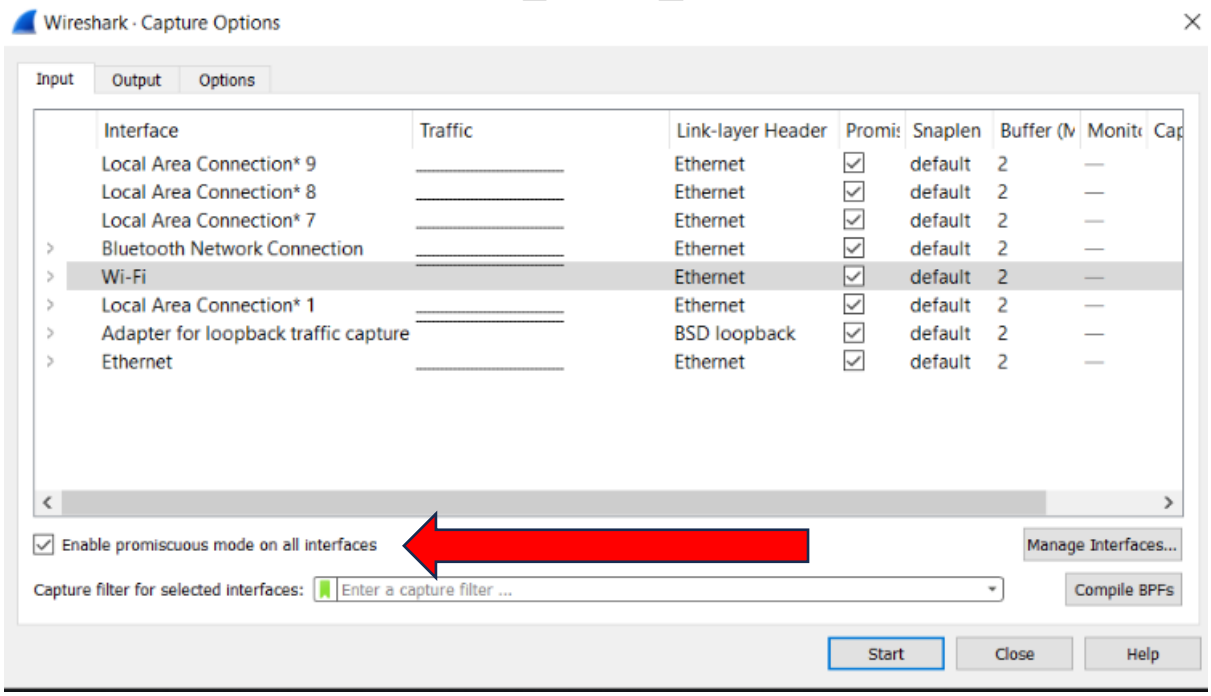


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Once you click the name of the network interface, you'll immediately observe packets appearing in real-time within Wireshark. Each packet sent to or from your system is captured and displayed for analysis.

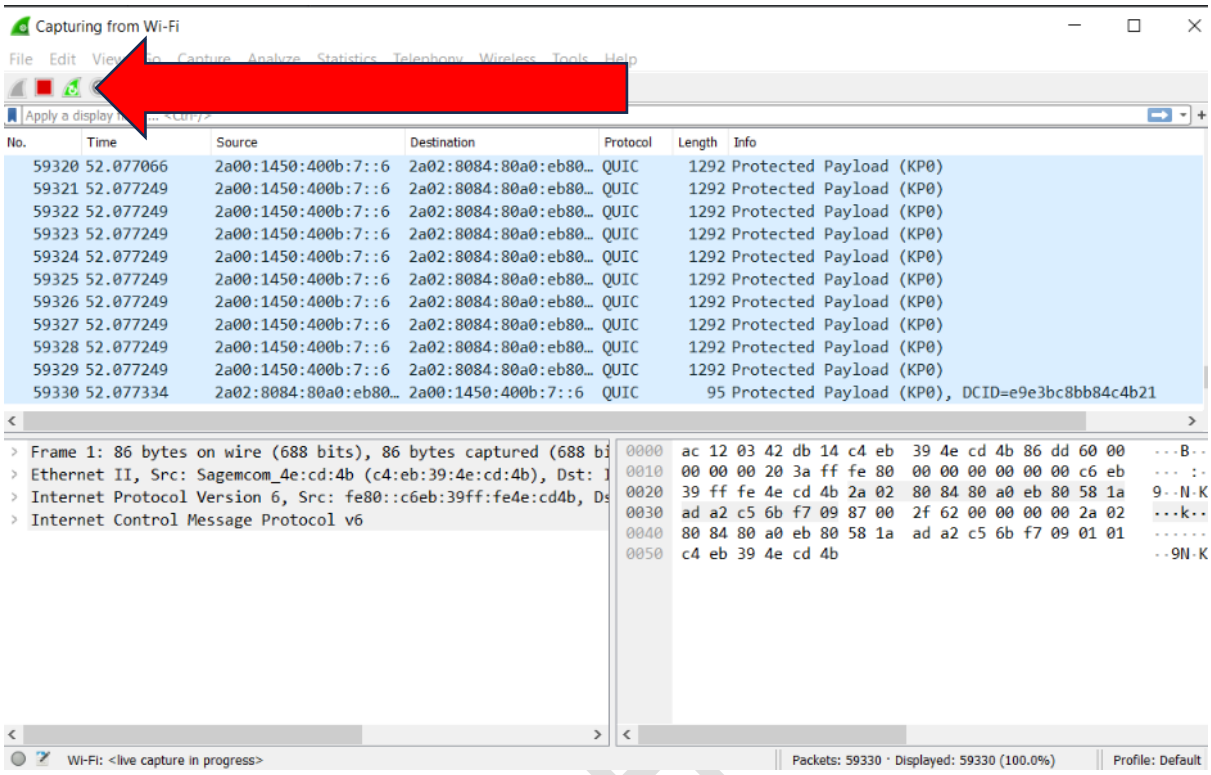
Wireshark, by default, captures not only the packets addressed to your network adapter but also all the other packets on the network. This means you can inspect all network traffic instead of just the traffic meant for your system. To confirm whether promiscuous mode is enabled, follow these steps:

1. Click "Capture" in the menu bar.
2. Select "Options."
3. In the "Capture Options" window/Input tab, verify that the "Enable promiscuous mode on all interfaces" checkbox is selected at the bottom of the window.



Enabling promiscuous mode allows Wireshark to capture all network traffic on the interface, providing a more comprehensive view of network activity for analysis.

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Click the red "Stop" button near the top left corner of the window when you want to stop capturing traffic.

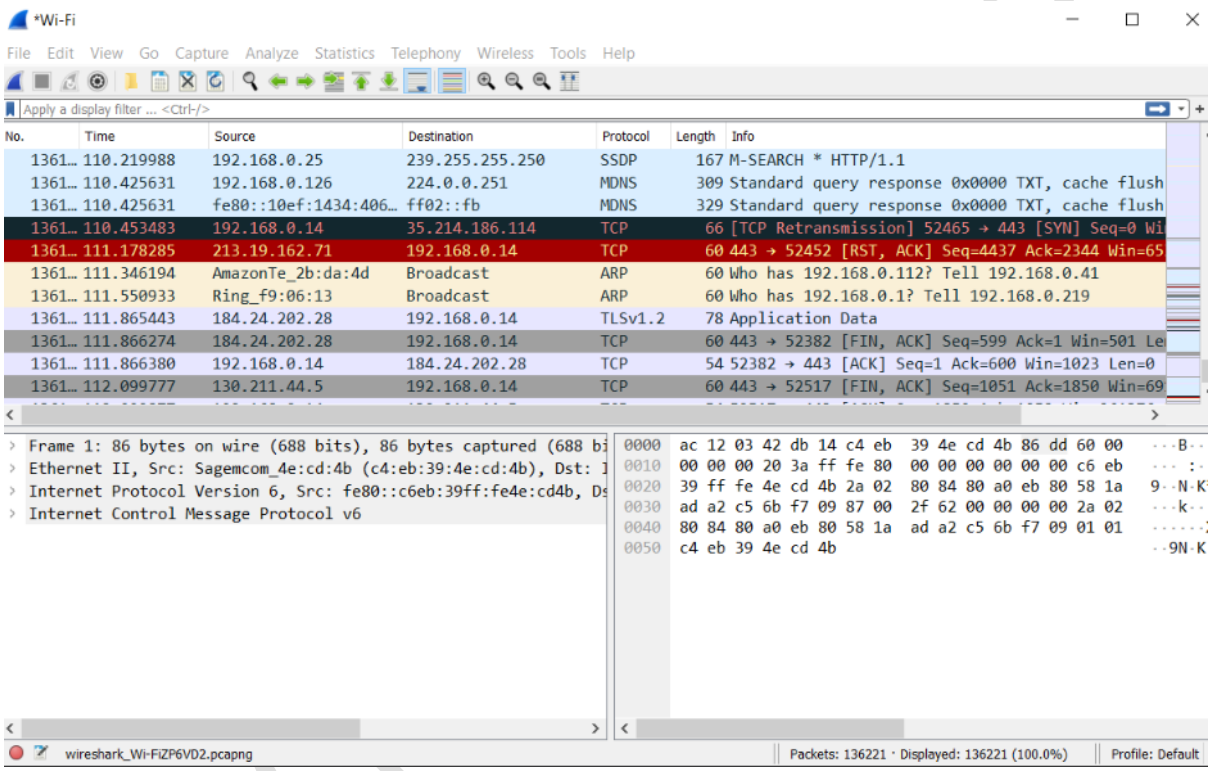
Wireshark employs a color-coding system to make it easier for users to identify the types of network traffic immediately. When you capture packets, you will notice different colours applied to them. Here's what some of the default colours represent:

- Light purple: This colour is assigned to TCP (Transmission Control Protocol) traffic. TCP is a common protocol for reliable data transmission.
- Light blue: UDP (User Datagram Protocol) traffic is denoted by this colour. UDP is a protocol for faster, connectionless data transfer.

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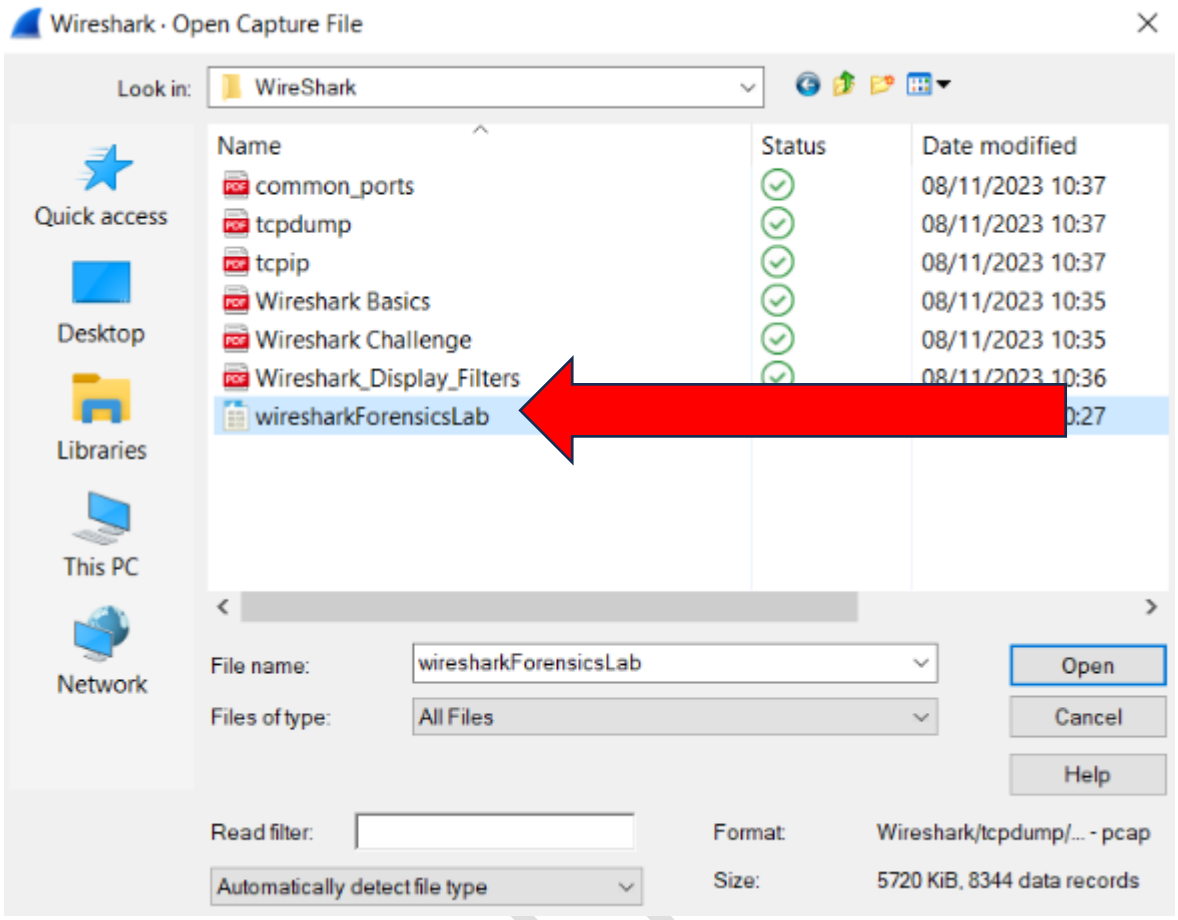
- **Black**: Packets marked in black indicate errors. These packets may have been delivered out of order or contain issues.

If you want to understand the meaning of other colour codes or customize the colouring rules to suit your preferences, you can access the "Colouring Rules" section by clicking on View > Colouring Rules. This feature allows you to modify and create rules for assigning colours to different types of traffic.



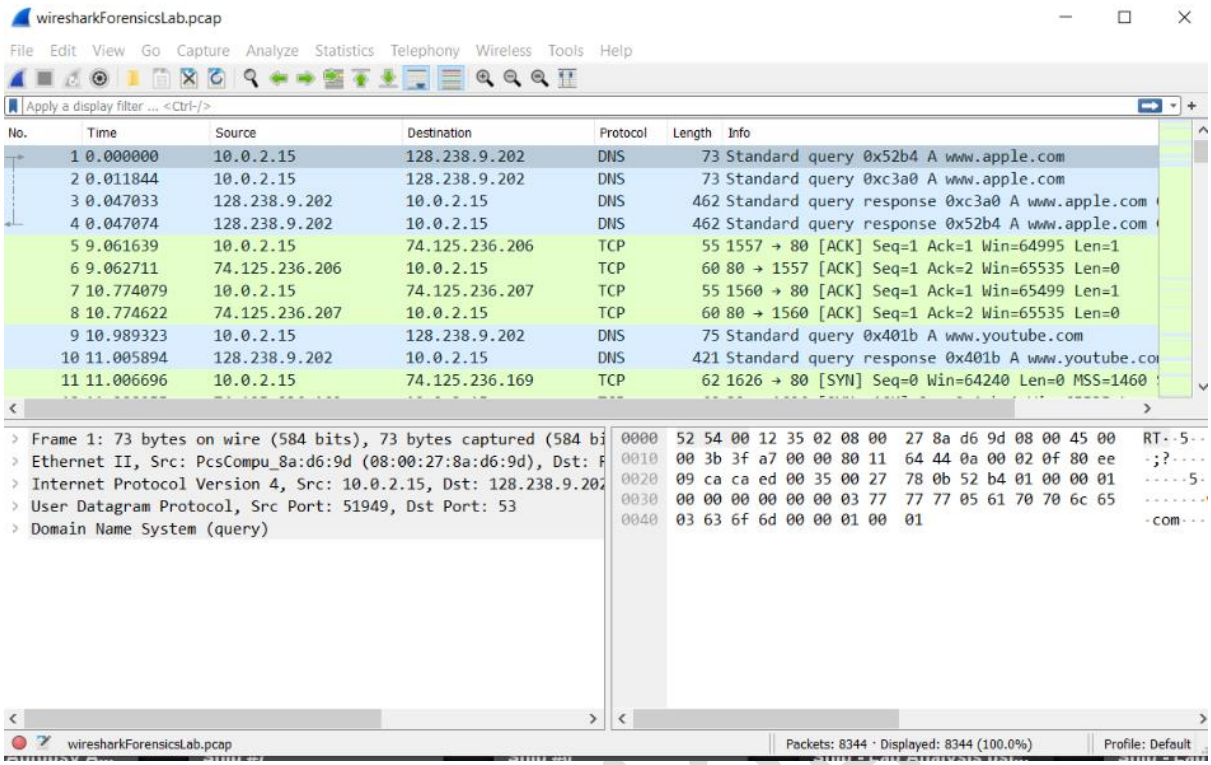
Click File/Open to upload the pcap from the folder you downloaded from Brighspace

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Wireshark Challenge

Welcome to the Wireshark Challenge! This exercise provides you with a sample pcap file and a set of questions to answer using Wireshark. Practice is key, so use this challenge to become more proficient with Wireshark.

Just like in any forensic activity, remember to record all your actions, methods, and solutions. This is a valuable practice and a good habit to develop. In future work-based scenarios, you will be required to submit these types of reports, so consider these challenges as essential preparation.

You will use the provided pcap file (ForensicsLab.pcap, available on Brightspace) to answer the following questions:

1. Google Searches: The user performed two different Google searches. What were the two search queries?

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2. File Downloads: The user downloaded a file from a site via FTP. What is the name of the downloaded file?
3. Email Activity: The user logged into an email account via the IMAP protocol and sent an email. Provide the following details: a. The user's email account login information b. The email addresses of the sender and recipient c. The key information contained in the email.
4. IRC Chat: What IRC chatroom did the user join?
5. Bonus: International Communications: This challenge can be completed using the "GeoIP" technology, which is implemented in newer versions of Wireshark. To tackle this bonus challenge, you'll need to:
    - Download the free GeoIP databases and configure Wireshark to access them.
        1. https://wiki.wireshark.org/HowToUseGeoIP.md
    - Enable GeoIP Lookups within Wireshark.
    - Use Firefox to view the map (other browsers may not display it correctly). Question for the bonus: Connections were established with servers located in three countries outside of America. What are the IP addresses of those servers, and in which countries are they located?

Remember to document your progress as you work through these questions. This challenge will help you enhance your network analysis skills, and it's essential to build a solid foundation for more complex forensic activities in the future.

https://www.wireshark.org/faq.html

Michael Hegarty